



(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

**VOICE CONTROL INTERACTIVE
LEARNING SYSTEM FOR CHILDREN
SY B.Tech. Project Implementation
Report (2307292)**

SUBMITTED BY

Shravani Kanse	202301090002
Aditi Gargade	202301090043
Bhakti Suryawanshi	202301070043
Aditi Patil	202301070136

**GUIDED BY
Prof. Vipin Yadav**

SCHOOL OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

MIT ACADEMY OF ENGINEERING, ALANDI (D), PUNE-412105

MAHARASHTRA (INDIA)

MAY, 2025

MIT

Academy of Engineering

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

CERTIFICATE

It is hereby certified that the work which is being presented in the SY B.Tech. Project Design Report entitled “ *VOICE CONTROL INTERACTIVE LEARNING SYSTEM FOR CHILDREN*”, in partial fulfillment of the requirements for the award of the **Bachelor of Technology in E&TC Engineering and Mechanical Engineering** and submitted to the **School of Electronics and Telecommunication Engineering of MIT Academy of Engineering, Alandi(D), Pune, Affiliated to Savitribai Phule Pune University (SPPU), Pune** is an authentic record of work carried out during an Academic Year 2020-2021, under the supervision of **Name of Advisor(s), School of Electronics and Telecommunication Engineering.**

Shravani Kanse	[202301090002]	[School of Mechanical Engineering]
Aditi Gargade	[202301090043]	[School of Mechanical Engineering]
Bhakti Suryawanshi	[202301070043]	[School of E&TC Engineering]
Aditi Patil	[202301070036]	[School of E&TC Engineering]

Date:

Wipin
Signature of Project Guide and Co-Guide

Project Guide/Co-Guide

School of E&TC Engineering,
MIT Academy of Engineering, Alandi(D), Pune

Signature of Dean

Dean

School of E&TC Engineering
MIT Academy of Engineering, Alandi(D), Pune

Signature of Internal examiner/s

Name

Affiliation

Signature of External examiner/s

Name *V. P. Rasel*

Affiliation *Comm. & I.T.*

SY Project Information-2024-25

A] Project Category		
01	Regular/Backlog/DSY	Regular
02	Interdisciplinary Project	Yes
	(If Interdisciplinary project Yes, Names of other schools of institute involved in project)	Mechanical School of Engineering
03	Industry Sponsored Project	No
	(If Yes, Name and address of Company)	-
	(Nature of Industry sponsorship)	-
04	Forefront area of project (WiCOM/SIP/EmSys/SysCon)	Yes, SIP
05	Project type (Purely Hardware/ Purely software/Hardware & Software)	Hardware and software
06	Sustainability Project (Project is related to Sustainability)	Yes
07	Social Project (Involvement of people from Society)	Yes
08	Hackathon Project (Problem statement from Hackathon)	No
B] Project Details		
09	Project to be continued till final year B.Tech	No
10	Project Guide (Name & School of Guide)	Dr. Vipin Yadav School of E&TC Engineering
11	Co-Guide (Name & School or Industry of Guide)	-
	(Industry Guide Contact Details-Email/Mobile)	-

CONTENTS

Acknowledgement		I
Abstract		II
List of Figures		III
List of Tables		IV
1.	Introduction	9
	1.1 Background	9
	1.2 Motivation	10
	1.3 Problem Statement	10
	1.4 Objective and scope	10
	1.5 Expected Results	10
2.	Literature Survey	12
	2.1 List of Referred Books and Papers	12
	2.2 Survey of Reference literature	13
3.	System Design	14
	3.1 Block diagram of Proposed System	14
	3.2 system parameters for expected outcome	15
	3.3 Hardware and Software Requirements	15
	3.4 Project Plan for Sem III and IV	17
4.	Design Verification with simulation	19
	4.1 Final Circuit diagram/Layout of Software	19
	4.2 Discussion on design improvements	22
5.	Conclusion	23

6.	References	24
7.	Appendix -1 Snap shots of Project Team at work/Results (Industry sponsored project add photos working industry)	25
8.	Appendix -2 (Pseudo code)	26
9.	Appendix -3 Datasheets	27
10.	Appendix -4 Reviews/Assignments and Activities	28
	10.1 Literature Review and Synopsis	
	10.2 Project Planning and Design	
	10.3 Power point presentation of Review-1	
	10.4 Power point presentation of Review-1	

ACKNOWLEDGEMENT

We would like to sincerely thank everyone who helped and supported us in completing this project, VOICE CONTROL INTERACTIVE LEARNING SYSTEM FOR CHILDREN This project has been a great learning experience, and we are very grateful to all those who guided and encouraged us during this journey.

First, we thank our project guide Prof Vipin Yadav, for his constant support, helpful feedback, and valuable advice. His guidance was a big reason why we were able to complete this project successfully.

We also want to express our gratitude towards respected School Dean Dr.Dipti Sakhare for her continuous encouragement.

We are also thankful to the School of E&TC Engineering at MIT Academy of Engineering, Pune, for providing the resources and facilities we needed to work on this project. The support from the institute was very important for us to achieve our goals.

We would be failing in our duty if we do not thank all the other staff and faculty members for their experienced advice and evergreen co-operation.

Name	Sign
-------------	-------------

Bhakti Suryawanshi	
---------------------------	--

Aditi Patil	
--------------------	--

Shravani Kanse	
-----------------------	--

Aditi Gargade	
----------------------	--

ABSTRACT

The "Voice Control Interactive Learning System for Children" project aims to revolutionize the way children engage with educational content by introducing an interactive and voice-activated learning tool. This system is designed to create an engaging and personalized learning experience using advanced speech recognition and natural language processing technologies. Children can interact with the system through simple voice commands, which are processed and responded to in real-time, fostering a dynamic and enjoyable educational environment.

The core of the system lies in its ability to understand and respond to children's queries, narrate stories, and provide tailored educational content through a user-friendly interface. By incorporating speech-to-text and text-to-speech functionalities, the system ensures seamless communication. This innovative approach not only simplifies learning but also promotes independent exploration, making it an invaluable tool for enhancing early childhood education.

List of figures

Sr.no	Figure Name	Page no.
Fig.1	Proposed Block Diagram	14
Fig.2	Proposed Block Diagram (b)	14
Fig. 3	Circuit Diagram	18
Fig.4	Flowchart	
Fig.5	Flowchart(b)	
Fig.6	VC02 (a)	25
Fig.7	VC02(b)	25
Fig.8	Model	26
Fig.9	Model(2)	26
Fig.10	Final Model(a)	27
Fig.11	Final Model(b)	27
Fig.12	3D Modeling(a)	28
Fig.13	3DModeling(b)	28
Fig.14	Gimini api link output	33
Fig.15	Screen output	36
Fig.16	Screen working output	36

List of Tables

Sr.no	Table Name	Page no.
Table. 1	Literature Survey	13
Table. 2	Sem III Project planning	16
Table. 3	Sem IV project planning	17

Chapter 1

Introduction

1.1 Background

The "Voice Control Interactive Learning System for Children" is a project that uses Artificial Intelligence (AI), Speech Recognition, and Natural Language Processing (NLP) to make learning fun and interactive. The system allows children to talk to it, play games, ask questions, and learn through stories and activities. It listens to their voice, understands what they say, and gives answers or instructions in a way that keeps them interested and engaged.

This system is designed to reduce the time children spend staring at screens by focusing on voice-based interactions. It encourages kids to explore and learn using their natural curiosity, helping them develop important skills while having fun. By making learning exciting and easy, the system supports children in discovering new things and enjoying the process of education.

1.2 Motivation

Helping young minds grow with smart technology! Our Voice Control Interactive Learning System makes education fun and engaging through interactive games, stories, and activities. It sparks curiosity, encourages creativity, and inspires children to learn at their own pace for a brighter future.

1.3 Problem Statement

To generate an interactive learning system for kids using voice control technology.

1.4 Objectives and

Scope Objectives:

- To make learning interactive and engaging for children.
- To simplify education through voice control technology.
- To provide a hands-free learning experience for children.
- To enhance accessibility in education for all children.
- To integrate fun and technology into the learning process

Scope:

- Interactive Learning Platform
- Gamified Learning Experiences
- Multi-Language
- Cultural Adaptability

TARGET AUDIENCE

- Working Parents
- Students
- Learning institutes
- Educational systems

1.5 Expected Results

Enhanced Safety for Linemen: The system is expected to provide a secure mechanism that prevents accidental power restoration during maintenance tasks, ensuring maximum safety for electrical linemen.

Password-Protected Access: Only authorized personnel with the correct password will be able to operate the circuit breaker, reducing the risk of unauthorized access.

Wireless Control via Android Application: The Android-based application will enable seamless, remote operation of the circuit breaker, making the process more efficient and user-friendly.

Reliability and Accuracy: The microcontroller-based system is anticipated to function reliably, with precise execution of commands to control the power supply.

Cost-Effective and Scalable Solution: The design is expected to be affordable and easily scalable, making it suitable for various applications in the power sector.

Reduced Human Errors: Automation of the circuit breaker operation will minimize manual intervention, thereby reducing the chances of operational errors.

Chapter 2

Literature Survey

2.1 List of Referred Books and Papers

Books:

- "Speech and Language Processing" by Daniel Jurafsky and James H. Martin
A comprehensive book on natural language processing and speech recognition, covering both theoretical and practical aspects.
- "Fundamentals of Speech Recognition" by Lawrence Rabiner and Biing-Hwang Juang
A classic text that provides a solid foundation in speech recognition, including the theory and algorithms used in speech processing systems.

Papers:

1. Electric Line Man Safety Using Password Based Circuit Breaker with Android App - IJRPR
2. Android App Based Circuit Breakers Controlling for Linemen Safety - ResearchGate

2.2 Survey of Reference literature

Table 2.1 Literature Survey

Sr. No.	Title	Authors	Summary	Key Features
1	Survey of Natural Language Processing for Education: Taxonomy, Systematic Review, and Future Trends	Yulia Sandilya, Sanjay Jha, Dipti Srinivasan	This paper surveys the various applications of NLP in education, categorizing them into areas like automated essay grading, sentiment analysis, and intelligent tutoring systems. It provides a systematic review of existing research and presents future trends and challenges, such as the need for more diverse data, the integration of multimodal learning, and the role of artificial intelligence in enhancing education through NLP.	Provides a taxonomy of natural language processing (NLP) applications in education. Systematic review of NLP in education research from 2010 to 2020. Identifies key challenges and opportunities for NLP in educational contexts.
2	A Systematic Literature Review on the Applications of Robots and Natural Language Processing in Education	V Maria Isabel Garcia-Segura, Marisa R. M. Amado, Pablo A. González-González	This paper reviews the applications of robots and NLP in education, emphasizing how robots can assist in educational contexts by using NLP for interaction with students. The study covers how robots enhance learning experiences, provide personalized tutoring, and foster engagement. It also addresses challenges such as designing robots that effectively communicate with students and the need for appropriate linguistic models in these systems.	M Focuses on the intersection of robots and NLP in educational settings. Discusses the impact of robots as educational tools and their integration with NLP. Examines both the theoretical and practical aspects of these technologies.

3	Natural Language Processing in Educational Research: The Evolution of Research Topics	David E. S. Peña, Manuel A. Zorrilla	This paper traces the evolution of NLP research in educational contexts, identifying key shifts in focus over three decades. The study highlights the growing interest in automated feedback systems, learning analytics, and virtual teaching assistants. It also discusses how advancements in NLP and machine learning are reshaping educational research, particularly in terms of personalized learning and large-scale educational data analysis.	Analyzes the evolution of research topics related to NLP in education. Focuses on the period from 1990 to 2020, tracking shifts in research interest. Highlights key technological advancements and areas of application.
4	Educational Chatbots for Project-Based Learning: Investigating Learning Outcomes for a Team-Based Design Course	Alessandro G. Martelloni, Paola Pazzaglia, Paola Zoppi	This paper explores the implementation of educational chatbots in a team-based project design course. The study investigates how chatbots, as conversational agents, can enhance the collaborative learning process, provide real-time support, and facilitate better interaction among students. The findings suggest that chatbots positively influence learning outcomes, particularly in terms of engagement and the development of problem-solving skills.	Investigates the use of educational chatbots in project-based learning environments. Focuses on a team-based design course and the impact on learning outcomes. Examines the role of chatbots in facilitating collaboration and knowledge sharing.

Chapter 3

System Design

3.1 Block diagram of Proposed System

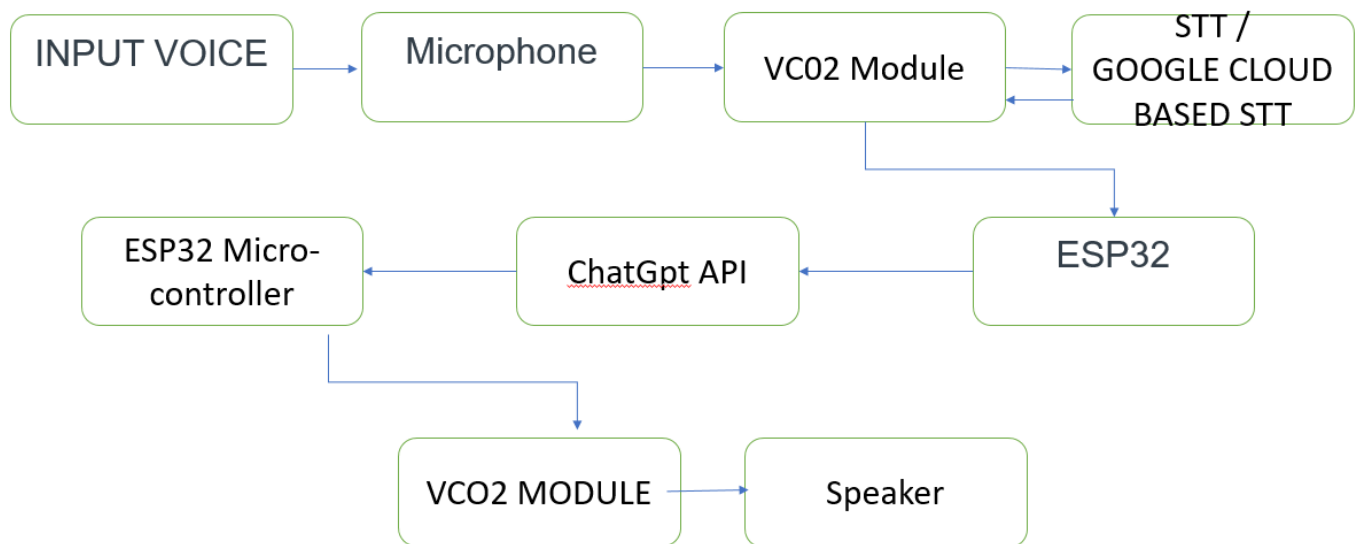


Figure 1 Flowchart

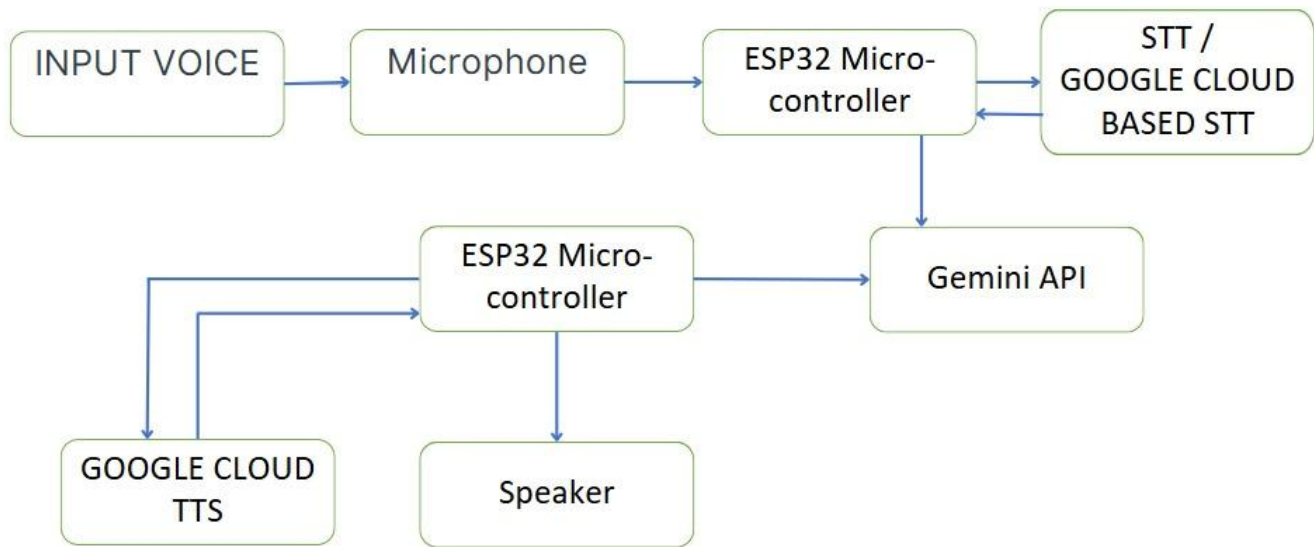


Figure 2 flowchart (b)

3.2 System parameters for expected outcome

- 3.1.1 Response Time
- 3.1.2 Latency in Communication
- 3.1.3 Current and Voltage Sensing Accuracy
- 3.1.4 Trip Time under Fault Conditions
- 3.1.5 Load Handling Capacity
- 3.1.6 Power Consumption of the System
- 3.1.7 Security and Authentication
- 3.1.8 User Interface Responsiveness
- 3.1.9 Reliability and Stability (Long-Term Operation)
- 3.1.10 Range and Stability of Wireless Communication
- 3.1.11 Alarm and Notification System
- 3.1.12 Fail-Safe Mechanism Testing
- 3.1.13 Temperature Monitoring and Management

3.3 Hardware and Software Requirements

Hardware:

- ESP32 Dev Board
- INMP441 MEMS Microphone
- MAX98357 I2S Amplifier

- 2W 8ohm Speaker
- TP4056 Battery Charging Module (MicroUSB)
- LiPo battery
- IR Proximity Sensor
- HT7833

Software:

- Arduino IDE
- ChatGPT API
- Voice.aithinker.com

3.4 Project Plan for Sem III and IV

Table 1

Activity	Details	Timeline
Project Ideation	Finalize the project idea, target audience (e.g., children aged 8–9 years), and scope.	Week 1 – Week 2
Literature Survey	Research related works and techniques (e.g., speech recognition, IoT integration).	Week 3 – Week 4
Component Procurement	Identify and procure necessary hardware components (e.g., ESP32, microphones, speakers).	Week 5
Basic Development	Develop basic functionalities like voice-to-text using ESP32 and integrate API responses.	Week 6 – Week 8
Prototype Assembly	Assemble an initial working prototype with basic features operational.	Week 9 – Week 10
Initial Testing	Test individual components like voice commands and text-to-speech features.	Week 11 – Week 12
Documentation	Prepare mid-term documentation of progress, challenges, and future steps.	Week 13 – Week 15

Table 2

Activity	Details	Timeline
Advanced Development	Add advanced features like interaction with ChatGPT API, offline processing, and safety modules.	Week 1 – Week 4
Integration	Combine all components into the working Brainy Robot prototype.	Week 5 – Week 6
User Testing	Conduct usability testing with children (target age group) and refine the system based on feedback.	Week 7 – Week 8
Performance Analysis	Evaluate system performance, accuracy, and response time.	Week 9
Final Refinement	Incorporate final changes based on performance analysis.	Week 10
Report Preparation	Document the entire project, including problem statement, methodology, and results.	Week 11 – Week 12
Presentation	Prepare for project viva, create slides, and practice presenting the final project.	Week 13 – Week 15

Chapter 4

Design Verification with simulation

4.1 Final Circuit diagram

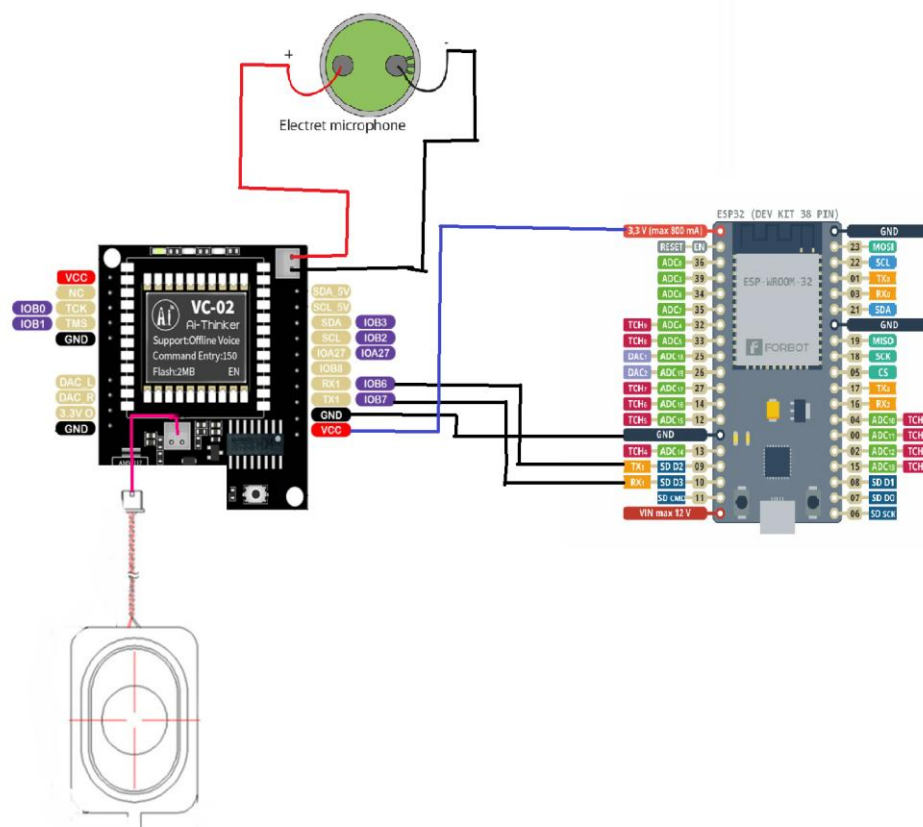


Figure 3

4.2 Discussion on design improvements

1. Voice Commands

- Teach the robot simple commands like "Tell me a story" or "What is 2+2?"

2. Fun Notifications

- Make the robot give voice alerts like "Time to play!" or "Good job on learning!"

3. Easy Power Setup

- Ensure the robot turns off automatically after 5 minutes of no use to save power.

4. Cute Design

- Add colorful stickers or shapes to make the robot look friendly and fun.

5. Simple Control App

- Create a basic app for parents where they can press buttons to guide the robot.

6. Play Songs or Stories

- Add a feature where the robot can play songs or tell short stories when asked.

7. Safety Features

- Ensure the robot has no sharp edges and works safely without overheating.

8. Easy Updates

- Add a button to update the robot's features when you learn something new to improve it.

Chapter 5

Conclusion

This project explored the application of voice processing technology, focusing on the conversion and recognition of voice signals. Through the implementation of advanced algorithms and signal processing techniques, we achieved accurate voice recognition, making it applicable in various real-world scenarios such as voice-controlled systems, security systems, and interactive applications.

Key findings include:

- The successful integration of various speech processing techniques for clear voice recognition.
- The ability to detect and distinguish different voice inputs with high accuracy.

Future Scope

- Future work will involve improving the model's robustness, expanding its vocabulary, and applying it to more complex real-world applications such as multilingual voice recognition systems.

Chapter 6

References

- a. ESP32. GitHub. by justin23456543.
- b. <https://github.com/justin23456543/ESP32>
- c. IoT Design Pro. (n.d.). ESP32 based Webserver for Text to Speech (TTS) Conversion. Retrieved August 22, 2024, from <https://iotdesignpro.com/projects/esp32-based-webserver-for-text-to-speech-tts-conversionnatural-language-processing> · GitHub Topics · GitHub
- d. WATElectronics. (n.d.). Audio Amplifier Circuit, Working and details of Nx, Lm386 Amplifiers. Retrieved August 19, 2024, from <https://watelectronics.com/audio-amplifier-circuit-working-and-details-of-nx-lm386-amplifiers>
- e. CodeBottle. (n.d.). codebottle.io. Retrieved August 16, 2024, from <https://codebottle.io>

Appendix -1

Snap shots of Project Team at work/Results

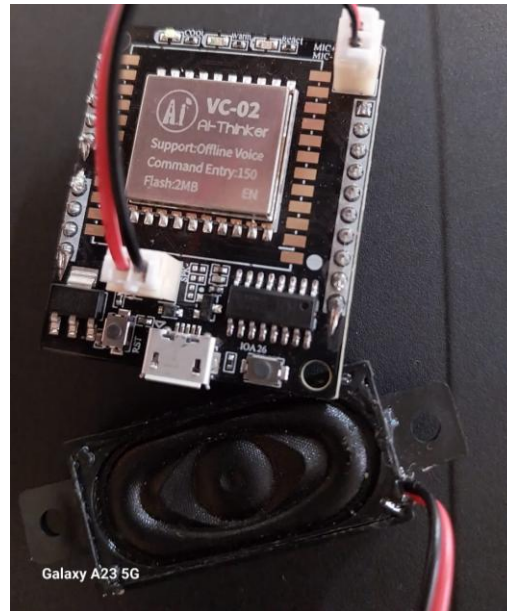


Figure 4

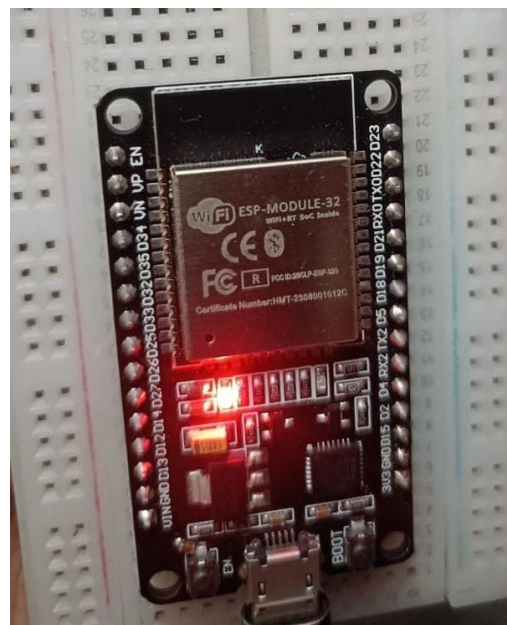


Figure 5

3D Modeling

Model 1

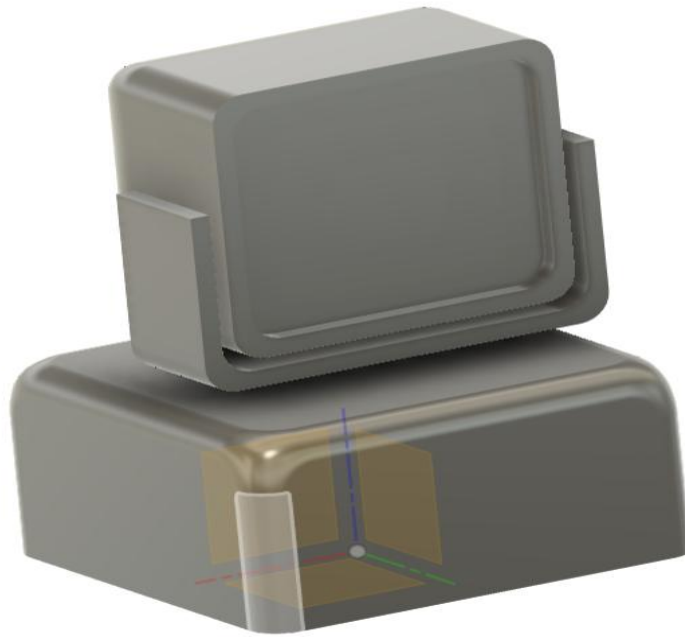


Figure 6 model

Model 2

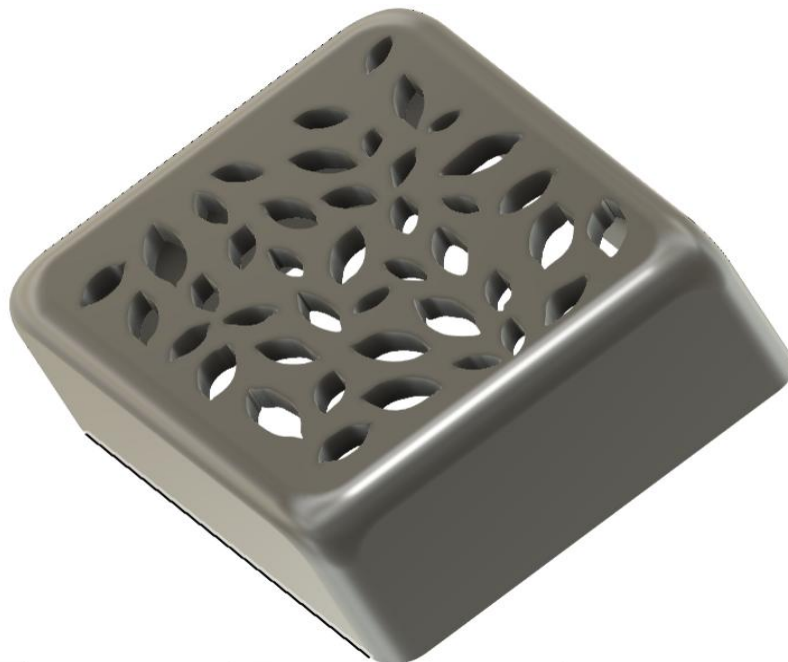


Figure 7 model(2)

Final Selected Model

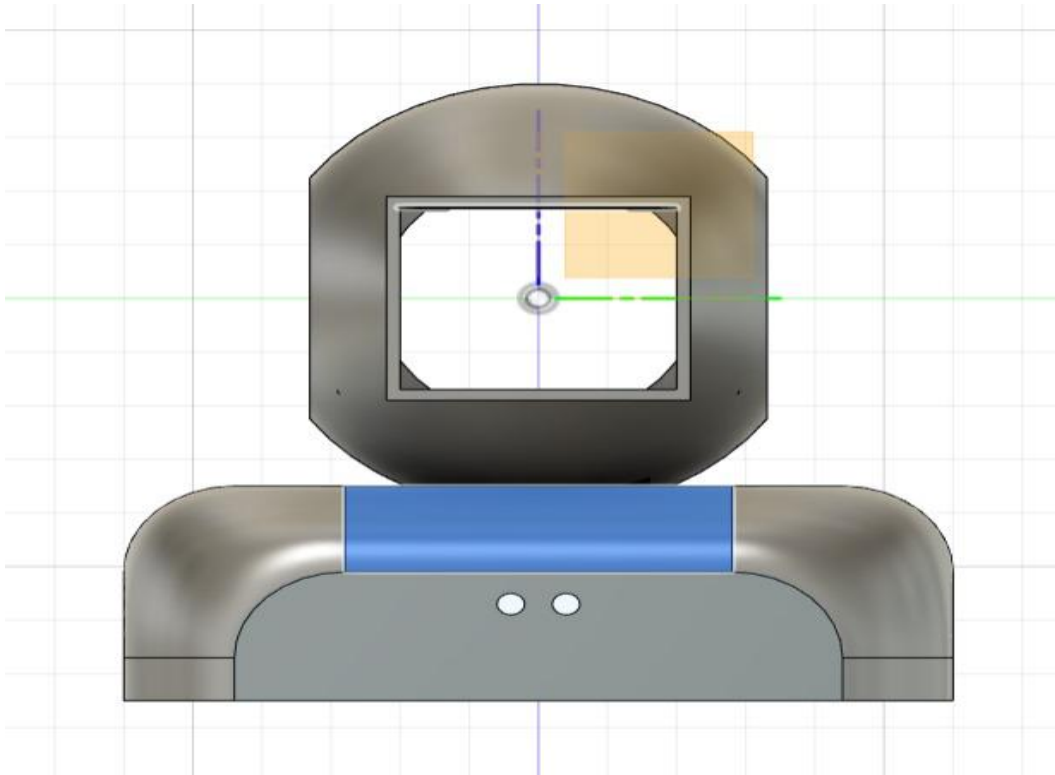


Figure 8 final model(a)

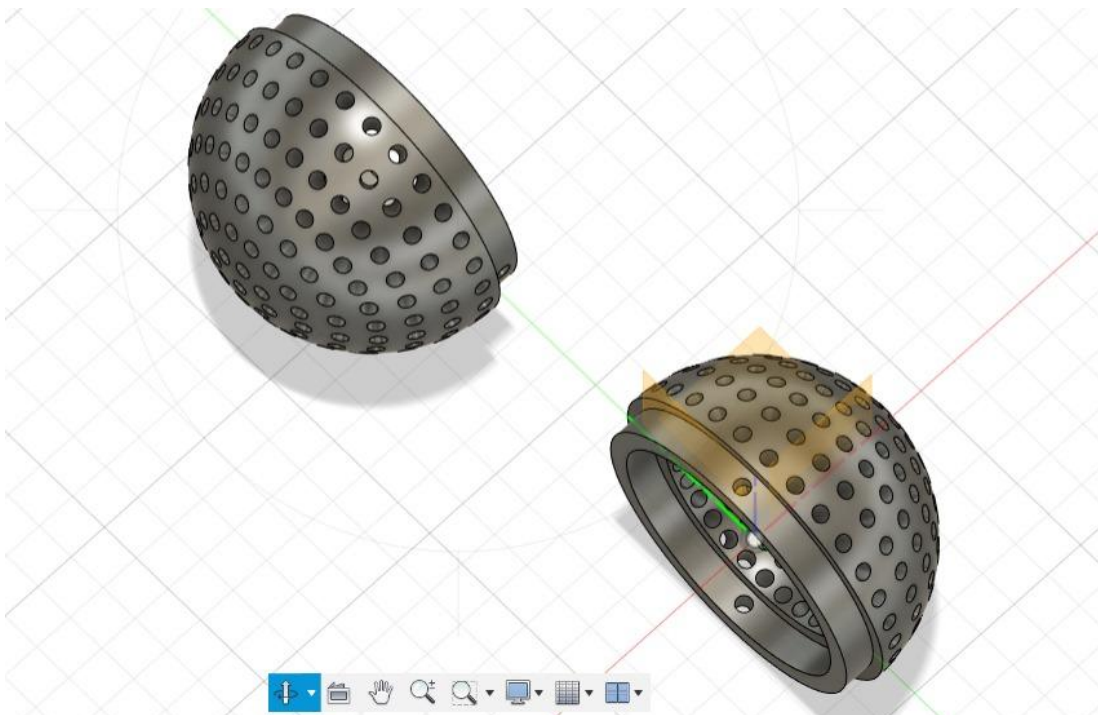


Figure 9 final model(b)

3D Modeling

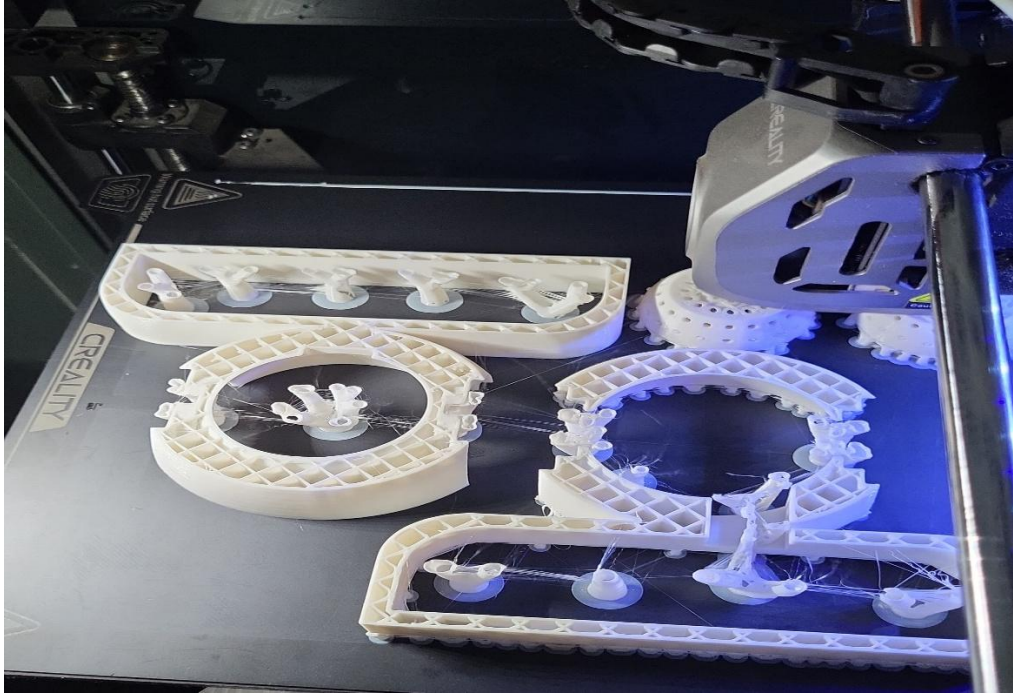


Figure 10 3D modeling (a)



Figure 11 3D Modeling(b)

Appendix - 2

Pseudo Code

Step 1: System Initialization

1. Power on the system.
2. Initialize the ESP32, microphone, and speaker.
3. Display “Ready for Command” on an LCD or app interface.

Step 2: Voice Command Input

4. Wait for a voice command from the user (e.g., “Tell me a joke” or “What is the weather?”).

Step 3: Voice-to-Text Conversion

5. Convert the spoken command into text using the ESP32.

Step 4: ChatGPT API Processing

6. Send the text command to the ChatGPT API for a response.
7. Receive and process the response text.

Step 5: Output Response

8. Convert the response text into speech using the ESP32.
9. Play the spoken response through the speaker.

Step 6: Reset

10. Wait for the next voice command or reset after inactivity.

Gemini API Link

```
#include <Arduino.h>

#include <WiFi.h>

#include <HTTPClient.h>

#include <ArduinoJson.h>

const char* ssid = "SSID";

const char* password = "PASS";

const char* Gemini_Token = "GEMINI_API_KEY";

const char* Gemini_Max_Tokens = "100";

String res = "";


void setup() {

    Serial.begin(115200);

    WiFi.mode(WIFI_STA)

;

    WiFi.disconnect();

    while (!Serial);

    // wait for WiFi connection

    WiFi.begin(ssid, password);

    Serial.print("Connecting to ");

    Serial.println(ssid);

    while (WiFi.status() != WL_CONNECTED) {

        delay(1000);
```

```
Serial.print(".");

}

Serial.println("connected");

Serial.print("IP address: ");

Serial.println(WiFi.localIP());

}

void loop()

{

Serial.println("");

Serial.println("Ask your Question : ");

while (!Serial.available())

;

while (Serial.available()) {

char add = Serial.read();

res = res + add;

delay(1);

}

int len = res.length();

res = res.substring(0, (len - 1));

res = "\"" + res + "\"";

Serial.println("");

Serial.print("Asking Your Question : ");
```

```

Serial.println(res);

HTTPClient https;

//Serial.print("[HTTPS] begin...\n");

if (https.begin("https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-
flash:generateContent?key=" + (String)Gemini_Token)) { // HTTPS

    https.addHeader("Content-Type", "application/json");

    String payload = String("{\"contents\": [{\"parts\": [{\"text\": \" + res + \"}], \"generationConfig\":
    {\"maxOutputTokens\": \" + (String)Gemini_Max_Tokens + \"}}");

    //Serial.print("[HTTPS] GET...\n");

    // start connection and send HTTP header

    int httpCode = https.POST(payload);

    // httpCode will be negative on error

    // file found at server

    if (httpCode == HTTP_CODE_OK || httpCode == HTTP_CODE_MOVED_PERMANENTLY) {

        String payload = https.getString();

        //Serial.println(payload);

        DynamicJsonDocument doc(1024);

        deserializeJson(doc, payload);

        String Answer = doc["candidates"][0]["content"]["parts"][0]["text"]

        // For Filtering our Special Characters, WhiteSpaces and NewLine Characters

        Answer.trim();

        String filteredAnswer = "";

        for (size_t i = 0; i < Answer.length(); i++) {

            char c = Answer[i];

            if (isalnum(c) || isspace(c)) {

```

```

filteredAnswer += c;

} else {

    filteredAnswer += ' ';

}

}

Answer = filteredAnswer;

Serial.println("");

Serial.println("Here is your Answer: ");

Serial.println("")

Serial.println(Answer);

} else {

Serial.printf("[HTTPS] GET... failed, error: %s\n", https.errorToString(httpCode).c_str());

}

https.end();

} else {

Serial.printf("[HTTPS] Unable to connect\n");

}

res = "";

}

```

Output

```

Ask your Question :
Asking Your Question : "who are you"
Here is your Answer:
I am a large language model trained by Google I am an AI that can process and generate human like text in response to a wide range of prompts and questions
Ask your Question :

```

Figure 12 gimini api link output

Screen Code

```
#include <WiFi.h>
#include <WiFiUdp.h>
#include <NTPClient.h>
#include <Adafruit_GFX.h>
#include <Adafruit_ST7735.h>

// TFT Pins
#define TFT_CS 15
#define TFT_RST 2
#define TFT_DC 4
#define TFT_SCLK 18
#define TFT_MOSI 23

// IR Sensor
#define IR_SENSOR 21

// Create TFT display object
Adafruit_ST7735 tft = Adafruit_ST7735(TFT_CS, TFT_DC, TFT_RST);

// WiFi credentials
const char* ssid = "AdSP";
const char* password = "17092005";

// Time client
WiFiUDP ntpUDP;
NTPClient timeClient(ntpUDP, "pool.ntp.org", 19800, 60000); // IST = UTC + 5:30

unsigned long lastChange = 0;
bool showTime = false;

void setup() {
  Serial.begin(115200);

  pinMode(IR_SENSOR, INPUT);

  tft.initR(INITR_BLACKTAB);
  tft.setRotation(1);
  tft.fillScreen(ST77XX_BLACK);
  tft.setTextColor(ST77XX_WHITE);
  tft.setTextSize(1);

  WiFi.begin(ssid, password);
```

```
Serial.print("Connecting to WiFi");

while (WiFi.status() != WL_CONNECTED) {
  delay(500);
  Serial.print(".");
}

Serial.println("Connected!");
timeClient.begin();
}

void loop() {
  timeClient.update();
  int irValue = digitalRead(IR_SENSOR);

  if (irValue == LOW) { // Object detected
    if (!showTime) {
      showTime = true;
      tft.fillScreen(ST77XX_BLACK);
      drawTimeAndDate();
    }
  } else {
    if (showTime) {
      showTime = false;
      tft.fillScreen(ST77XX_BLACK);
      drawAnimatedEyes();
    }
  }

  delay(1000); // Adjust based on responsiveness needed
}

void drawTimeAndDate() {
  tft.setCursor(10, 20);
  tft.setTextSize(2);
  tft.setTextColor(ST77XX_GREEN);
  tft.print(timeClient.getFormattedTime());

  // Simulate Date (NTPClient doesn't provide date directly)
  tft.setCursor(10, 50);
  tft.setTextSize(1);
  tft.setTextColor(ST77XX_CYAN);
  tft.print("Date: use RTC or API");
}
```

```
tft.setCursor(10, 70);  
tft.print("Day: Mon");  
}
```

```
void drawAnimatedEyes() {  
  // Simple static "happy" eyes example  
  tft.fillCircle(40, 40, 15, ST77XX_WHITE);  
  tft.fillCircle(40, 40, 5, ST77XX_BLACK);  
  
  tft.fillCircle(100, 40, 15, ST77XX_WHITE);  
  tft.fillCircle(100, 40, 5, ST77XX_BLACK);  
  
  // Add sleep or other expressions by changing shape/colors  
}
```



Figure13 screen output

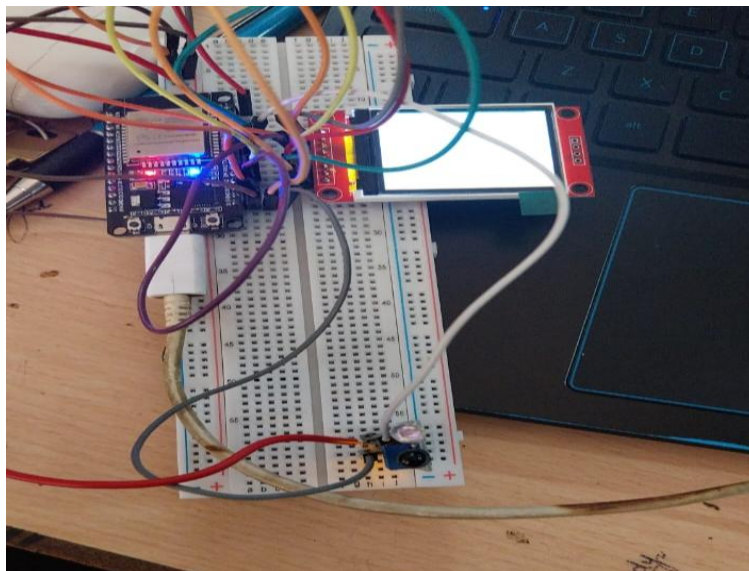


Figure 14 screen working model

Appendix - 3

Datasheets

Product Overview

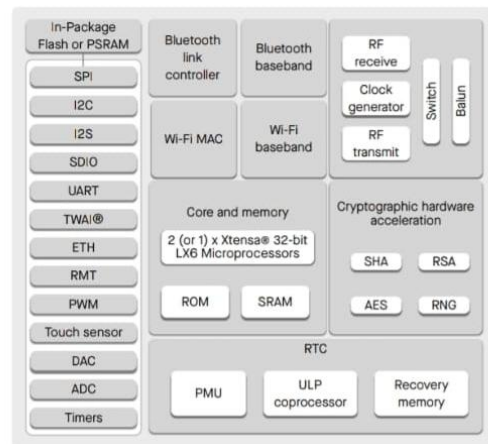
ESP32 is a single 2.4 GHz Wi-Fi-and-Bluetooth combo chip designed with the TSMC low-power 40 nm technology. It is designed to achieve the best power and RF performance, showing robustness, versatility and reliability in a wide variety of applications and power scenarios.

The ESP32 series of chips includes ESP32-D0WD-V3, ESP32-D0WDR2-V3, ESP32-U4WDH, ESP32-S0WD (NRND), ESP32-D0WDQ6-V3 (NRND), ESP32-D0WD (NRND), and ESP32-D0WDQ6 (NRND), among which,

- ESP32-S0WD (NRND), ESP32-D0WD (NRND), and ESP32-D0WDQ6 (NRND) are based on chip revision v1 or chip revision v1.1.
- ESP32-D0WD-V3, ESP32-D0WDR2-V3, ESP32-U4WDH, and ESP32-D0WDQ6-V3 (NRND) are based on chip revision v3.0 or chip revision v3.1.

For details on part numbers and ordering information, please refer to Section 1 ESP32 Series Comparison. For details on chip revisions, please refer to [ESP32 Chip Revision v3.0 User Guide](#) and [ESP32 Series SoC Errata](#).

The functional block diagram of the SoC is shown below.



ESP32 Functional Block Diagram

1. Product overview

VC-02 is a low-cost pure offline speech recognition module developed by Shenzhen Ai-Thinker Technology Co., Ltd. The module uses the voice chip US516P6 launched by Unisound, and continuously optimizes and innovates algorithms in speech recognition technology. The offline recognition algorithm and chip architecture are deeply integrated to provide customers with ultra-low-cost offline speech recognition solutions. It can be widely and quickly applied to smart home, all kinds of smart small household appliances, 86 boxes, toys, lamps and other products that need voice control.

The US516P6 chip uses a 32-bit RSIC architecture core and incorporates a DSP instruction set specifically for signal processing and speech recognition, an FPU arithmetic unit that supports floating-point operations, and an FFT accelerator.

VC-02 supports offline identification of 150 local instructions, RTOS lightweight system, rich peripheral interfaces, and simple and friendly custom tools.

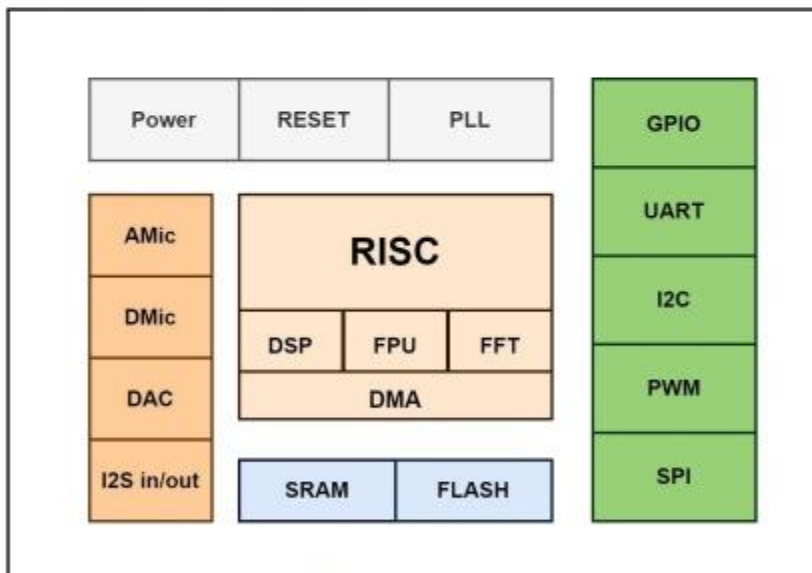


Figure 1 Main chip architecture diagram

10 Appendix 4

10.1 Literature Review and synopsis

Voice Recognition Technology

- **Research Paper Title:** "A Comparative Study of Speech Recognition Techniques"
- **Summary:**

"This paper compares different ways to convert speech into text, which is exactly what we're doing with the VC02 module in my project. The paper talks about how some systems, like the one we're using, process voice commands offline, meaning they don't need internet access to understand basic commands. That's helpful because it ensures faster responses and works well when there's no internet. It's a beneficial educational tools where kids need quick answers without waiting for long processing times."

AI in Education Systems

Research Paper Title: "Artificial Intelligence in Education: Promises and Challenges"

Summary:

"This paper is all about how AI is changing the way we teach and learn. It highlights how virtual assistants and AI chatbots can help students by answering their questions and guiding them through lessons—exactly what my project is doing. In our case, the ChatGPT API acts as the 'brain' of the robot, helping kids get answers to their questions in a conversational way. It shows how tools like these can make learning more interactive and personalized."

10.2 Project Planning and Design

Input and Processing:

- **Microphone:** A highly sensitive microphone that converts analogue signal to digital signal. It is typically used in applications requiring high-quality audio capture.
- **ESP32 Micro-controller:** A powerful microcontroller with built-in Wi-Fi and

Bluetooth capabilities. It serves as the brain of the robot, handling tasks such as processing audio data, communicating with other components, and executing the overall control logic.

- **STT/Google Cloud Based STT:** A cloud-based speech-to-text service that uses advanced algorithms to transcribe spoken language into text. It leverages machine learning expertise to achieve high accuracy.
- **NLP:** A field of artificial intelligence that deals with the interaction between computers and human (natural) languages. In this context, NLP algorithms are used to understand the meaning and context of the transcribed text, enabling the robot to respond appropriately.

Output Generation and Delivery:

- **Google Cloud TTS:** A cloud-based text-to-speech service that converts text into human-like speech. It utilizes sophisticated voice synthesis techniques to produce natural-sounding audio.
- **VCO2 Module:** A specialized hardware or software component that can modify the synthesized speech to enhance its quality, naturalness, or expressiveness. It may involve techniques like speaker customization, noise reduction, or audio effects.
- **Speaker :** A loudspeaker that converts the amplified electrical signal into sound waves, producing the audible output

10.3 Video Link

https://drive.google.com/drive/folders/1aAR0xuZudtPpdbxi_apmhaFu9zsqVtb3